Fitbuddy AI

Project Development Phase Documentation & Testing Framework

PROJECT PHASE	TARGET SYSTEM	TESTING DATE	STATUS
Phase 5 — Dev	GenAI & ML Engine	September 2026	Approved / UAT

1. Functional & Performance Testing Plan

1.1 Objective & Scope

Application Under Test: Fitbuddy AI Engine (LLM API Integrations, Dynamic Workout/Nutrition Recommendation Algorithms, Computer Vision Motion/Form Tracking).

Scope: Real-time query response times, visual exercise tracking latency, token generation speed, payload stress testing, and recommendation accuracy.

1.2 Performance & Reliability Criteria

METRIC / COMPONENT	TARGET BENCHMARK	THRESHOLD LIMIT	TEST ENVIRONMENT
LLM Response Time (TTFT)	< 1.2 seconds	> 2.5 seconds (Fail)	Staging / API Gateway
Computer Vision Frame Processing	≥ 30 FPS	< 18 FPS (Fail)	Edge / On-Device Testing
Concurrent Active Users	10,000 users	15,000 peak	AWS Load Generator
API Failure Rate	< 0.1%	> 0.5% (Fail)	Production-Simulated Payload

1.3 Key Functional Test Cases

- **Dynamic Meal Plan Generation:** Validate macronutrient calculation and diet preference handling (e.g., vegan, keto, allergies).
- **Pose Estimation & Form Feedback:** Verify joint-angle tracking accuracy during core movements (e.g., squats, pushups, deadlifts) with live camera feeds.
- **Conversational AI Fitness Coach:** Ensure contextual memory retention across multi-turn user interactions.

2. User Acceptance Testing (UAT) Report

2.1 Executive Overview

Project Name: Fitbuddy AI — Personal Fitness & Nutrition Companion | **UAT Phase:** Phase 5 Review

MODULE / FEATURE	TOTAL TEST CASES	PASSED	FAILED	PENDING FIX
User Onboarding & Profiling	15	15	0	0
AI Workout Planner	22	21	1	0
Real-time Pose Detection	18	16	2	2 (Camera Latency)
Macronutrient Camera Scanner	12	12	0	0
Progress Dashboard & Analytics	10	10	0	0

> **Key Feedback & Remediation Plan:**
> • **Pose Detection Latency:** Low-end mobile devices experienced frame drops during real-time rep counting. *Resolution:* Downgrade neural network model resolution on lower-tier hardware automatically.
> • **Meal Plan Adjustments:** Immediate caloric scaling based on workout completion has been optimized.

3. Business Intelligence & Analytics Integration

3.1 Key Performance Indicators (KPIs) Captured

User Engagement: Daily Active Users (DAU), Monthly Active Users (MAU), average session length.
Workout Completion Rate: Percentage of AI-generated workout plans completed to end.
Model Inference Metrics: Average token usage per session, prompt latency distribution, and cost per active user.

3.2 Integration Architecture

